Software Engineering

BSCS-6G

Assignment # __1_____

Submitted by:

i- Aleena Qayyum L23-0981
ii- Aleena Orazi L23-0820
iii-Eman Aamir l23-0612
iv-Muskan Irfan L23-1022
v-Armish Rao l23-0504

Department of Computer Science, School of Computing

Fast NUCES, Lahore

Project Proposal

Title: Hirely

1. Problem Statement

High Volume of Applications

Organizations often receive hundreds of applications for a single job or internship. Many applicants do not meet the required skills or experience, making manual screening time-consuming and inefficient. This delays the hiring process and increases recruiter workload.

Lack of Candidate Transparency

Candidates frequently apply without fully understanding job requirements. They rarely receive clear feedback about their eligibility or rejection reasons, leading to frustration, confusion, and repeated irrelevant applications.

Inefficient Communication

Existing recruitment systems often fail to provide real-time updates on application status. Recruiters spend extra time responding to emails and queries, while candidates remain uncertain about their progress.

Difficulty in Skill Matching

Employers struggle to quickly identify suitable candidates because resumes may not clearly highlight relevant skills. This results in longer hiring cycles and sometimes poor recruitment decisions.

Need for a Solution

The proposed system introduces a **Hirely** combined with a two-sided recruitment and service platform.
It pre-screens candidates, reduces irrelevant applications, improves transparency, enables AI-

based CV filtering, and allows companies to act as both employers and service providers, making the system efficient, scalable, and practical for real-world use.

2. Project Scope

User Roles (Actors)

The system supports **two main sides**:

1) Employer / Company Side

Companies act as **employers** and **service providers**.

Account & Company Profile

- Register and login
- Create and manage company profile (mini-website)
- Add:
    - Company description
    - Logo and banner
    - Contact details
    - Services offered
    - Products (if any)
    - Portfolio / past work

Job & Internship Management

- Post jobs and internships with:
    - Job title
    - Required skills
    - Experience level
    - Location
    - Internship / Job type
    - Remote / On-site
    - Application deadline
- Edit, close, or delete job postings

Applicant Management

- View list of applicants
- Download and review CVs
- Shortlist candidates
- Update application status:
    - Accepted
    - Rejected

    - Interview scheduled

Service & Product Management

- Add and edit services offered
- Add product listings
- Allow users to:
    - View services
    - Inquire or request services
    - Contact the company

2) Candidate / Student / User Side

Users act as **job seekers** and **service consumers**.

Account & Profile Management

- Register and login
- Create personal profile
- Add:
    - Skills
    - Education
    - Experience
- Upload CV / resume

Job & Internship Search

- Browse job and internship listings
- Filter by:
    - Skills
    - Location
    - Internship / Job
    - Remote / On-site

Applications & Tracking

- Apply to jobs and internships
- Track application status in real time:
    - Pending
    - Shortlisted
    - Interview
    - Rejected

Hirely

- Eligibility check before applying

- Skill-match validation
- Awareness questions related to job requirements
- Highlight missing skills and gaps

Service Marketplace

- Browse company profiles (mini-websites)
- View services and products offered
- Search services by category
- Contact companies
- Request or book services

Notifications

- Receive notifications for:
    - New job/internship postings
    - Application status updates
    - Interview schedules
    - New services launched by companies

Exclusions & Assumptions

Exclusions

- Payroll management
- Background verification

Assumptions

- Users provide accurate information
- System is used in a controlled organizational environment

3. Tools and Technologies

Frontend – React

React will be used to develop a responsive and interactive user interface. It enables:

- Smooth navigation
- Real-time status updates
- Dynamic dashboards for companies and users

Backend – Node.js

Node.js handles:

- Authentication and authorization
- Application workflows
- Job postings and applicant management
- API integration
  Its asynchronous nature supports scalability and high traffic.

Database – SQL Server

SQL Server stores:

- User and company profiles
- Job and internship data
- Applications and status history
- Services and products
  Its relational structure ensures secure and efficient data management.

4. Review of Similar Existing Projects

Popular platforms such as **LinkedIn**, **Indeed**, **Glassdoor**, and **Upwork** provide recruitment or service-based features.

Strengths

- Large user base
- Job discovery and networking
- Resume uploads and recruiter access

Weaknesses

- No pre-application eligibility checks
- High number of irrelevant applications
- Limited transparency and feedback
- Recruitment and services are handled separately

5. What Makes This Project Different

Hirely (Key Innovation)

Unlike traditional systems, this platform **filters candidates before application submission**.
It ensures:

- Eligibility verification
- Skill-gap awareness
- Reduction in spam applications

Two-Sided Platform

Companies can:

- Hire talent
- Offer services and products

  Users can:
- Apply for jobs
- Act as clients for company services

Improvement Over Existing Systems

- Shifts filtering from post-application to pre-application
- Saves recruiter time
- Improves candidate quality
- Provides transparency and early feedback

Real-World Impact

- Faster hiring
- Fairer recruitment
- Better candidate experience
- Practical for startups, SMEs, and service-based companies